

Customer Behavior Analysis

A structured retail dataset used to uncover purchase patterns, optimize marketing strategies, and increase customer lifetime value through Python EDA, SQL analysis, and Power BI dashboards.



About the Dataset

CUSTOMER	EMAIL	DATE	AMOUNT
[REDACTED]	[REDACTED]	20141114 13:51	\$260
[REDACTED]	[REDACTED]	20121116 13:51	\$150
[REDACTED]	[REDACTED]	20141116 13:51	\$920
[REDACTED]	[REDACTED]	20141116 13:51	\$75

Tracks every customer interaction: what they buy, how often, how much they spend, and whether they respond to discounts.

Type

Retail / E-commerce, Structured (Tabular), Customer Transaction Level

Numerical Fields

Age, Purchase Amount, Previous Purchases

Categorical Fields

Category, Payment Method, Season



BUSINESS CONTEXT

Why This Data Matters



High-Value Customers

Identify top spenders and frequent buyers



Purchase Patterns

Understand what, when, and how often customers buy



Discount Optimization

Measure promo code effectiveness



Marketing & CLV

Improve campaigns and customer lifetime value

Breaking Down the Business Problem

Core Question: How can a retail company understand customer purchasing behavior to increase revenue, improve retention, and optimize marketing?

Low Retention

Customers not purchasing frequently — need to identify one-time vs. frequent buyers

Ineffective Campaigns

Do discounts increase purchases? Which customers respond to promo codes?

Product & Payment Insights

Which categories perform best? Which payment methods are preferred?

Seasonal & Segmentation

Does season impact sales? Identify high-value, frequent, and discount-sensitive segments

Dataset Columns

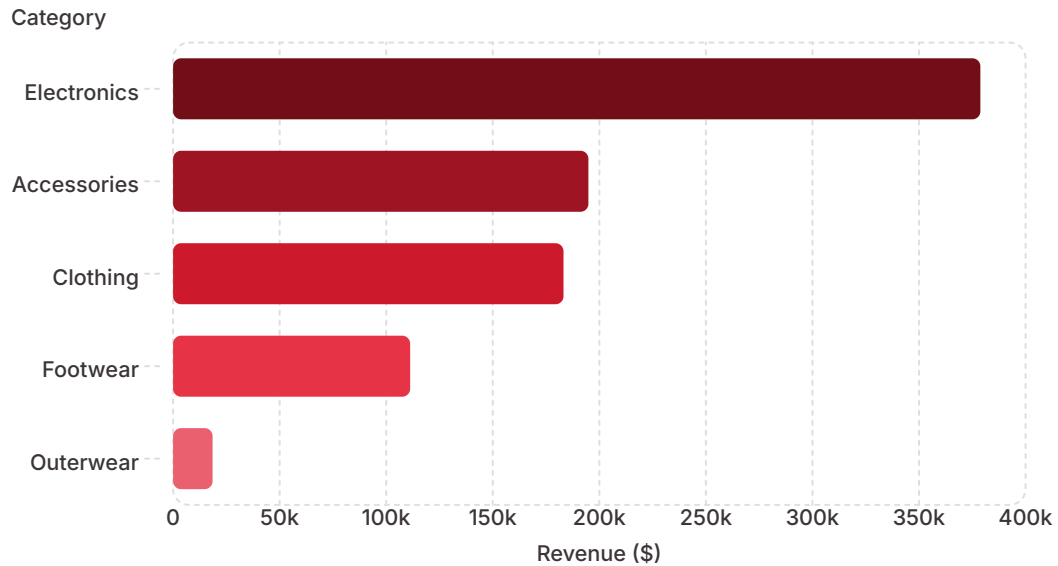
Customer ID	Unique identifier
Age, Gender	Age; Male / Female / Other
Item Purchased	Specific product
Category	Clothing, Electronics, etc.
Purchase Amount	Amount spent
Location	Geographic location
Size, Color	Product size (S/M/L); color
Season	Winter, Summer, etc.
Review Rating	Customer product rating
Subscription Status	Yes / No
Shipping Type	Shipping method selected
Discount Applied	Whether discount was applied
Previous Purchases	Number of past purchases
Payment Method	Mode of payment
Frequency of Purchases	Weekly, Monthly, etc.

Data Preparation in Python



Key cleaning steps: category mapping for consistency, category-specific missing value logic (e.g., Electronics → "Not Applicable"), product-level mean for Review Rating, zero for Previous Purchases, and duplicate removal for accurate reporting.

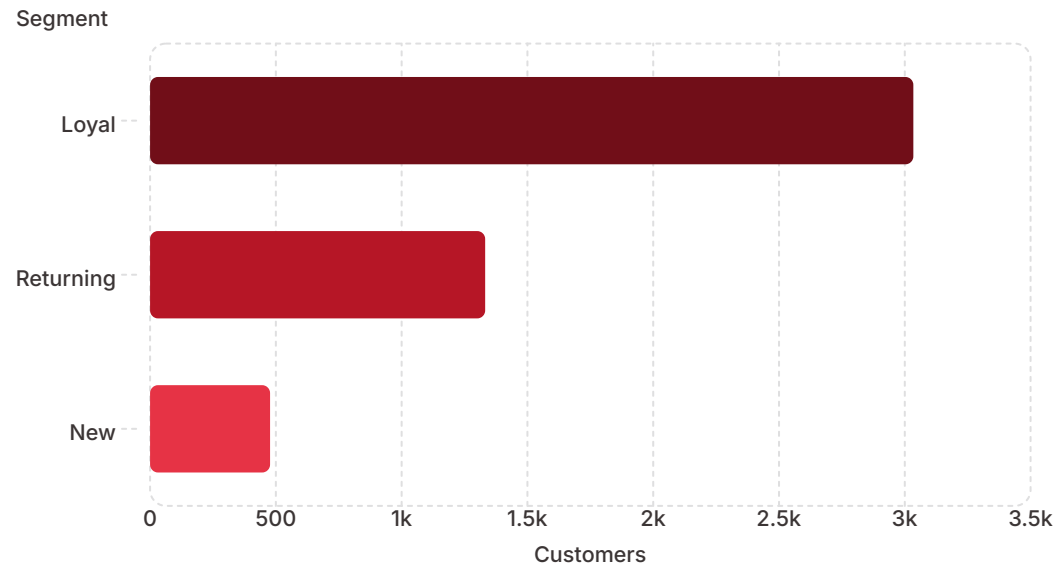
Revenue & Category Insights



Key Findings

- **Electronics** leads revenue at \$378,785 — nearly double Accessories
- Discounts raise avg. purchase to **\$202.73** vs. \$167.16 without
- **Subscribers** spend \$250.99 avg. vs. \$152.33 for non-subscribers
- Male: \$358,295 · Female: \$304,964 · Other: \$223,340

Customer Segments & Behavior



Segmentation Insights

- **Loyal Customers** (3,034) — highest previous purchase count
- Repeat buyers: **70% unsubscribed**, 30% subscribed
- Age 51+ drives highest revenue: **\$354,427**
- Top discount-used items: Laptop (51.75%), Phone (51.46%)

Dashboard Highlights

5K

Total Customers

29

Unique Items

3.63

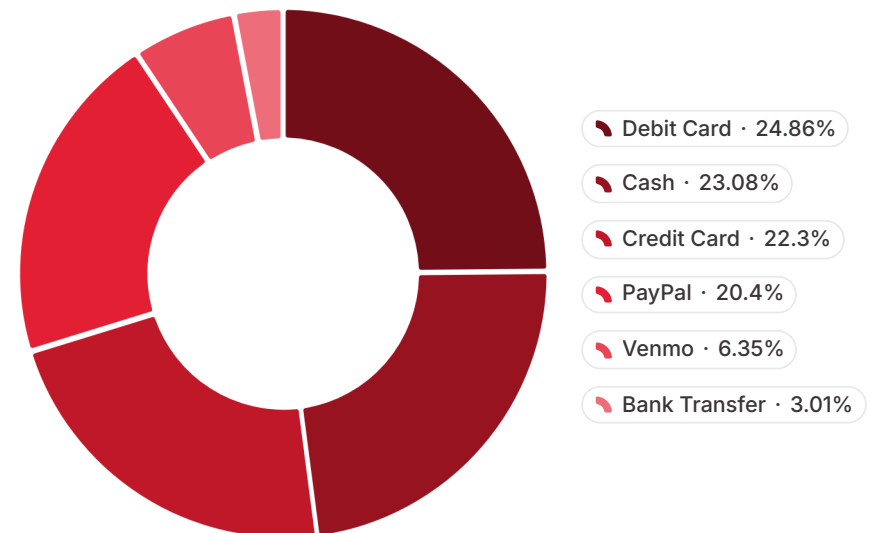
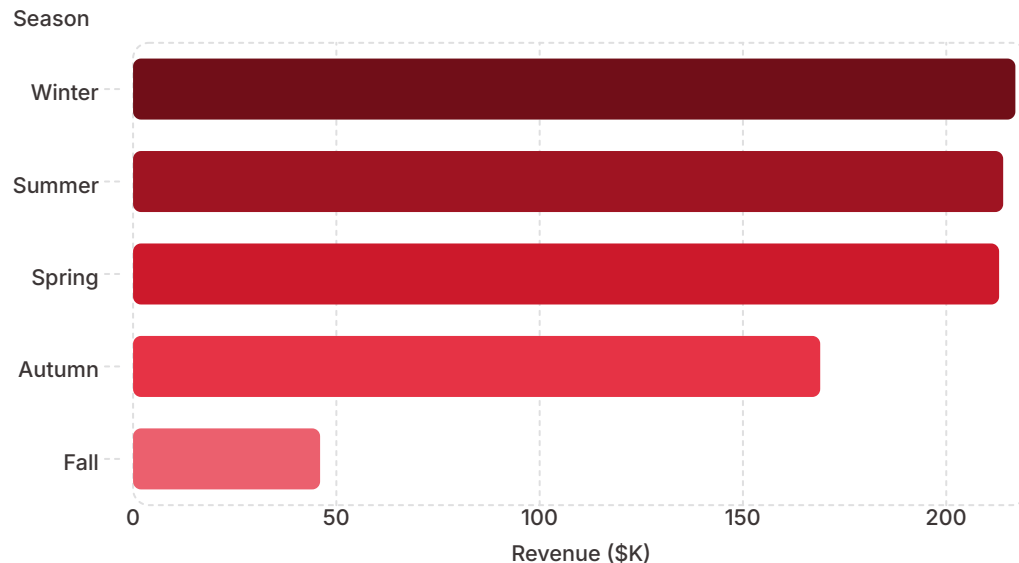
Avg Review

886K

Total Spend (\$)

183

Avg Spend (\$)



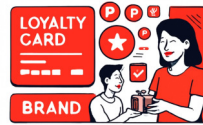
KEY TAKEAWAYS

Top Insights & Recommendations



Electronics Dominates

Highest revenue category —
prioritize inventory and
promotions here



Loyalty Drives Value

3,034 loyal customers;
subscription correlates with
higher spend



Seasonal Strategy

Winter, Summer, Spring drive
73% of revenue — plan
campaigns accordingly



Payment Preferences

Debit, Cash, Credit Card lead
— optimize checkout for
these methods